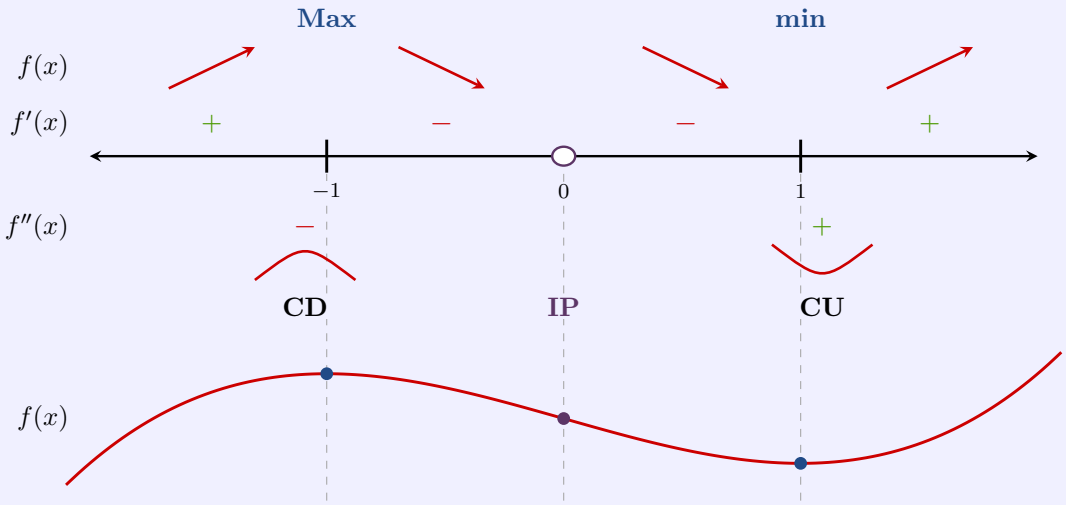
L2.3 How Derivatives Affect the Shape of a Graph

The derivatives table

① Find the **CNs** ($f' = 0$ or DNE) and the **PIPs** ($f'' = 0$ or DNE).
② Mark them all on **one** number line.
③ Plug a test value from each piece into f' and f''. Write the signs.
④ Draw arrows above (↗ or ↘) and cups below (⌒ or ⌣).
⑤ Sketch f underneath, lined up with the number line.

Ex. $f(x) = x^3 - 3x$. $f' = 3x^2 - 3 = 0$ at $x = \pm 1$ (CNs). $f'' = 6x = 0$ at $x = 0$ (PIP).



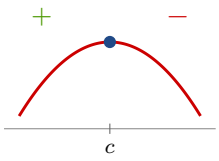
CN critical number · **CP** critical point · **PIP** possible inflection point · **IP** inflection point · **CU** concave up · **CD** concave down · **HT** horizontal tangent · **@** at

THE THREE TESTS

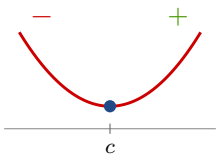
I/D Test. If $f'(x) > 0$, $f \uparrow$. If $f'(x) < 0$, $f \downarrow$.

1st Derivative Test. At a critical number c — where $f'(c) = 0$ *or* $f'(c)$ DNE:

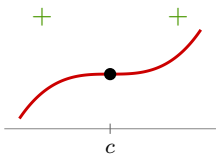
f' : $\oplus \to \ominus$ local Max $\quad \ominus \to \oplus$ local min $\quad \ominus \to \ominus$ or $\oplus \to \oplus$ no local extremum



$\oplus \to \ominus$: local Max



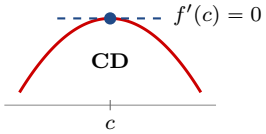
$\ominus \to \oplus$: local min



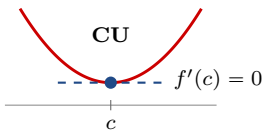
$\oplus \to \oplus$: no local extremum

2nd Derivative Test. **CU** if $f'' > 0$. **CD** if $f'' < 0$. **IP** if CU $\to$ CD or CD $\to$ CU.

$f'(c) = 0$ and CD $\Rightarrow$ local Max $\quad f'(c) = 0$ and CU $\Rightarrow$ local min



$f'(c) = 0$ and CD: local Max

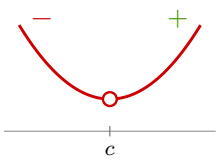


$f'(c) = 0$ and CU: local min

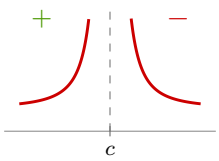
THE TWO CONDITIONS FOR A LOCAL EXTREMUM

For a local min or Max at a critical number c:

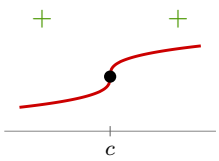
① the point must **exist** @ c $\quad$ ② f' must **switch sign** across c



① fails: hole @ c



① fails: VA @ c



② fails: $x^{1/3} + 3$, no switch